

Sadik Bera Yuksel

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PROFESSIONAL SUMMARY

Ph.D. candidate in Computer Engineering specializing in machine learning, with a focus on reinforcement learning and autonomous systems. Experienced in developing and deploying learning-based algorithms for robotic decision-making and control in complex environments. Proficient in Python and modern ML frameworks such as PyTorch, with strong experience in building, training, and evaluating machine learning models for real-world applications, as well as managing research and technical projects.

EDUCATION

Ph.D. in Computer Engineering Expected May 2028
Northeastern University, Boston, MA | GPA: 4.0/4.0

M.S. in Computer Engineering May 2025
Northeastern University, Boston, MA | GPA: 4.0/4.0

Concentration: Computer Vision, Machine Learning, and Algorithms

Relevant Coursework: Robotic Sensing and Navigation, Reinforcement Learning, Mobile Robotics, Advances in Deep Learning, Verifiable Machine Learning, Formal Methods for Dynamical Systems

Bilkent University June 2023
B.S. in Electrical and Electronics Engineering

Relevant Coursework: Introduction to Machine Learning, Microprocessors, Natural Language Processing

Korea University February - June 2022
Study Abroad Program, Electrical and Electronics Engineering

Relevant Coursework: Statistical Modeling for Deep Learning, Computer System Design, Artificial Intelligence

PROFESSIONAL EXPERIENCE

Northeastern University Boston, MA
Doctoral Researcher (Advisor: Prof. Derya Aksaray) September 2023 - Present

- Conducted research on safe autonomy by integrating Signal Temporal Logic into reinforcement learning and robotic foundation models for constrained decision-making under spatio-temporal tasks.
- Trained and evaluated deep reinforcement learning policies using PyTorch and adapted them within a foundation-model-based framework for robust action selection under temporal logic specifications.
- Developed a temporal-logic-guided motion planning framework with contingency constraints to ensure robust autonomous navigation under uncertainty with probabilistic guarantees.
- Deployed the planning algorithm on a real drone in a GPS-denied environment, integrating perception, state estimation, control, and runtime monitoring for autonomous task execution.

Northeastern University Boston, MA
Teaching Assistant January - May 2026

- Teaching Assistant for EECE5644: Introduction to Machine Learning and Pattern Recognition, assisting with instruction, grading, and student support for machine learning theory and applications.

HAVELSAN Ankara, Turkey
Robotics and Autonomous Systems Intern June - August 2021

- Implemented autonomous navigation for a differential drive robot in a simulation environment using ROS and Python.
- Developed a vision-based face recognition system on Qualcomm Robotics RB5 platform using OpenCV.

PUBLICATIONS

- **S.B. Yüksel** and D. Aksaray, “Specification-Aware Distribution Shaping for Robotics Foundation Models”, under review at *International Conference on Intelligent Robots and Systems (IROS)*, 2026. arXiv:2603.17969
- **S.B. Yüksel**, A.T. Buyukkocak and D. Aksaray, “Shielded Reinforcement Learning Under Dynamic Temporal Logic Constraints”, to appear in *American Control Conference (ACC)*, 2026.
- X. Lin, **S.B. Yüksel**, Y. Yazıcıoğlu, and D. Aksaray, “Probabilistic Satisfaction of Temporal Logic Constraints in Reinforcement Learning via Adaptive Policy-Switching”, Proc. of the Learning for Dynamics & Control (L4DC), 2025.
- **S.B. Yüksel**, A. Taheri, Y. Yazıcıoğlu, and D. Aksaray, “Robust Temporal Logic Planning Under Contingency Constraints”, International Conference on Automation Science and Engineering (CASE), 2025.

SKILLS

Programming Languages	Python, C++, MATLAB, Assembly
Robotics & Control	Reinforcement Learning, Motion Planning, State Estimation, Control Systems, SLAM, Sensor Fusion, MuJoCo, ROS
Machine Learning & AI	PyTorch, JAX, CUDA, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV
Tools and Software	Linux, Git, Docker, LTSpice, Vivado, Diptrace

ACADEMIC PROJECTS

Sketch-to-Image Translation for Human Identification Spring 2025
Northeastern University

- Designed and trained a Generative Adversarial Network (GAN) model in PyTorch for image generation from sketches, yielding a 28.2% increase in cosine similarity and a 3.5% improvement in SSIM over the baseline, which indicates enhanced feature-level and structural similarity to ground truth images.

Real-Time Localization and Mapping for Autonomous Vehicles Fall 2023
Northeastern University

- Developed and deployed a real-time visual-inertial SLAM pipeline using ORB-SLAM3 in ROS, fusing camera and IMU measurements for robust localization and mapping in driving environments using real vehicle data.

Indoors Drone Localization Using UWB Sensors Spring 2023
Bilkent University

- Developed an indoor drone localization system in Python that fuses Ultra-Wideband (UWB) distance data with IMU measurements using trilateration and filtering algorithms to determine accurate real-time position estimates, achieving an average localization error of 0.26 meters in real time.
- Deployed the algorithm on the drone’s onboard computer and created a real-time monitoring interface for ground-based visualization with an update rate of 20 Hz.

Vision-Language Model for Image Understanding Spring 2023
Bilkent University

- Trained LSTM and GRU-based models with attention mechanisms using PyTorch to generate image captions using the Flickr8k and Flickr30k datasets, achieving a BLEU score of 0.5168
- Optimized model performance through hyperparameter tuning with the Optuna library in Python and compared Greedy Search and Beam Search caption generation methods to evaluate caption quality.